

National Service Training Program (NSTP) Management System for Davao del Norte State College



**A Capstone Project Presented to the
Faculty of the Institute of Computing
Davao del Norte State College
New Visayas, Panabo City**

Project Team

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Chapter 1

INTRODUCTION

A. Background of the Study

National service is more than just “serving the public”, it involves serving under specific political and economic systems, promoting nationalism, and helping to build that nationalism. Various countries around the world implement national service programs, these programs can be military, non-military, or compulsory, and some are linked to the educational system, such as the National Service Training Program (NSTP) in the Philippines [1]. Research has shown various impacts of service learning on students in different countries. These studies generally found that service learning affects students academically, civically, and socially. The NSTP mandated by Republic Act No. 9163 (the NSTP law) aims to develop civic consciousness and defense preparedness among students by promoting service ethics and instilling nationalism and patriotism. The NSTP has three components: (1) Civic Welfare Training Service (CWTS), (2) Literacy Training Service (LTS), (3) Reserve Officers’ Training Corps (ROTC). These components are designed to increase youth participation in national well-being.

The Civic Welfare Training Service (CWTS) focuses on community service activities in health, education, environment, safety, and morality, while the Literacy Training Service (LTS) trains students to teach literacy and numeracy to children and out-of-school youth. The Reserve Officers’ Training Corps (ROTC) requires male and female students to undergo military training, whereas all students must participate in the National Service Training Program (NSTP) for one academic year. Schools and colleges are adopting Student Information Systems (SIS) to store, organize, and access student information. Studies show that computerized students help reduce manual errors, improve the data accuracy, and speed up record processing [2]. They also enhance the monitoring of student academic status and institutional performance [3] [4].

Digital systems are also being implemented in many colleges and universities in the Philippines to improve and enhance student record management. Nevertheless, there are still offices where manual procedures and processes are used that lead to delays, inconsistency of data and inability to create reports. In the office of the National Service Training Program (NSTP), at Davao del Norte State College, student status reports, certificate documents, and schedule of activities are not yet integrated into a centralized system. This causes efficiency, monitoring, and data management gaps. Based on the current local situation, there is a clear

need to develop an Information Management System for NSTP students to improve accuracy, speed, and performance of the coordinator. Therefore, this study proposes the development of an NSTP Management System for Davao del Norte State College that will centralize student records, automate certificate generation, and streamline reporting and monitoring processes, ultimately enhancing the efficiency and effectiveness of NSTP operations.

B. Company Overview

The National Service Training Program was also established in Davao del Norte State College, where it was officially implemented on November 13, 2009, to increase the rate of student involvement in community service and nation-building initiatives. It is a mandatory course for first year college students, which is categorized into three sections, which include: Literacy Training Service (LTS), Civic Welfare Training Service (CWTS), and Reserve Officers Training Corps (ROTC). It has a head NSTP Coordinator, Razzy Atowan, who is in charge of LTS and CWTS, whereas the ROTC aspect is led by Mark Anthony Librea with (8) LTS instructors, (14) CWTS instructors and, (30) student officers in the ROTC as assistants to execute the program. The NSTP program has 393 students under LTS, 1,245 students under CWTS and 1,030 students under ROTC as of 2025. The NSTP Office is charged with the responsibility of planning, organizing, implementing and monitoring all the related activities including conducting outreach programs, literacy programs, environmental programs, disaster preparedness training and leadership.

Figure 1. Below is the Organizational Structure showing the National Service Training Program in Davao del Norte State College. At the top is the Vice President for Academic Affairs Girley S. Gumanao, PHD who provides overall academic leadership and supervision for the program. Under her is the NSTP Coordinator, Mrs. Razul A. Atowan, who manages and oversees the implementation of the CWTS and LTS programs, The team also includes the NSTP instructors, who assist in delivering and facilitating the program components . Alongside her is the ROTC coordinator, Mr. Mark Anthony G. Librea who is responsible for managing and organizing all ROTC activities in the school. Under his supervision are the ROTC commanders, who lead and oversee.



**NATIONAL SERVICE TRAINING PROGRAM
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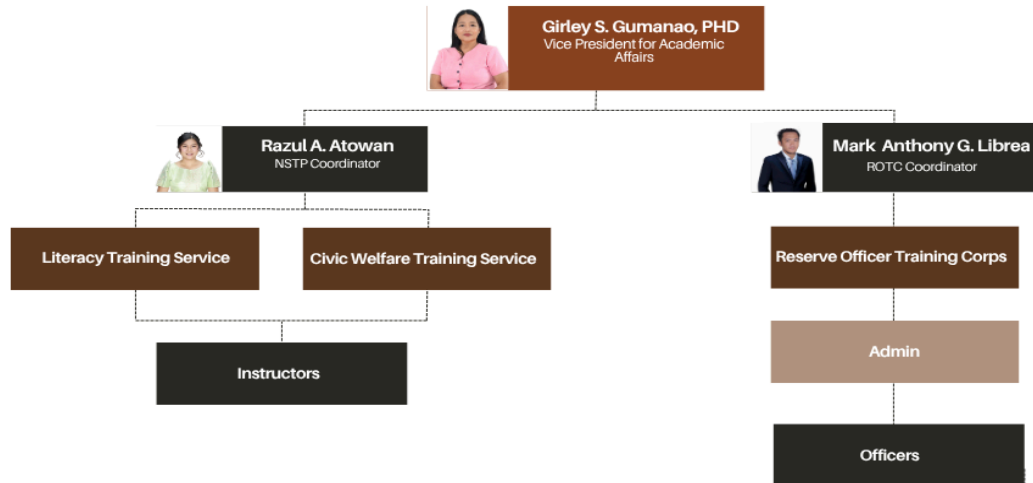


Figure 1: Organizational Structure of National Service Training Program (NSTP)

C. Problems Encountered

The lack of operational efficiency in the National Service Training Program (NSTP) office in the Davao del Norte State College is being caused by the fact that it continues to use manual and semi-digital systems to handle student records, instructor submissions, grading and report generation.

The following are the identified problems in the NSTP Office:

- **Manual Student Information Encoding:** Student information is entered manually in the NSTP office. This situation increases the chances of data duplication, misplaced records, and delays in retrieving student information.

- **Difficulty in Locating Student Grades:** Student grades are checked manually to determine whether the students passed the NSTP. This situation makes it difficult and time-consuming to locate and verify student grades.
- **Lack of Centralized Program and Section Management:** There is no centralized system to manage NSTP components, sections, and it is difficult for the NSTP Coordinator to organize CWTS, ROTC, LTS programs and control semester-specific settings.
- **Limited Monitoring of Instructor Submissions and Activities:** Instructor submissions such as activity plans, accomplishment reports, and grading progress are not monitored in real time and do not follow a centralized process. This causes delays in tracking instructor activities and slows down the process.
- **Manual Generation of Certificates and Reports:** Certificates and reports are prepared manually by the NSTP office. This consumes a significant amount of administrative time and increases the risk of errors in serial numbers and student information.

D.Existing Business Process

Figure 2, illustrates the manual workflow for the NSTP enrollment, grading, and certification process managed by the Coordinator. The process begins with the Coordinator logging into the system and manually downloading student files and instructor class lists from the IAIMS. These lists are then opened in Google Sheets, where the Coordinator performs a manual cross-check against a Master List of all students; those who match are included in the final list, while others are excluded. After validation, the Coordinator manually classifies students into their respective NSTP components and organizes the encoded data. Once grades are inserted, the Coordinator sorts the students into Pass or Fail categories and sends the necessary information to CHED via email. Upon receiving the official serial numbers from CHED, the Coordinator manually creates the completion certificates, marking the end of the process.

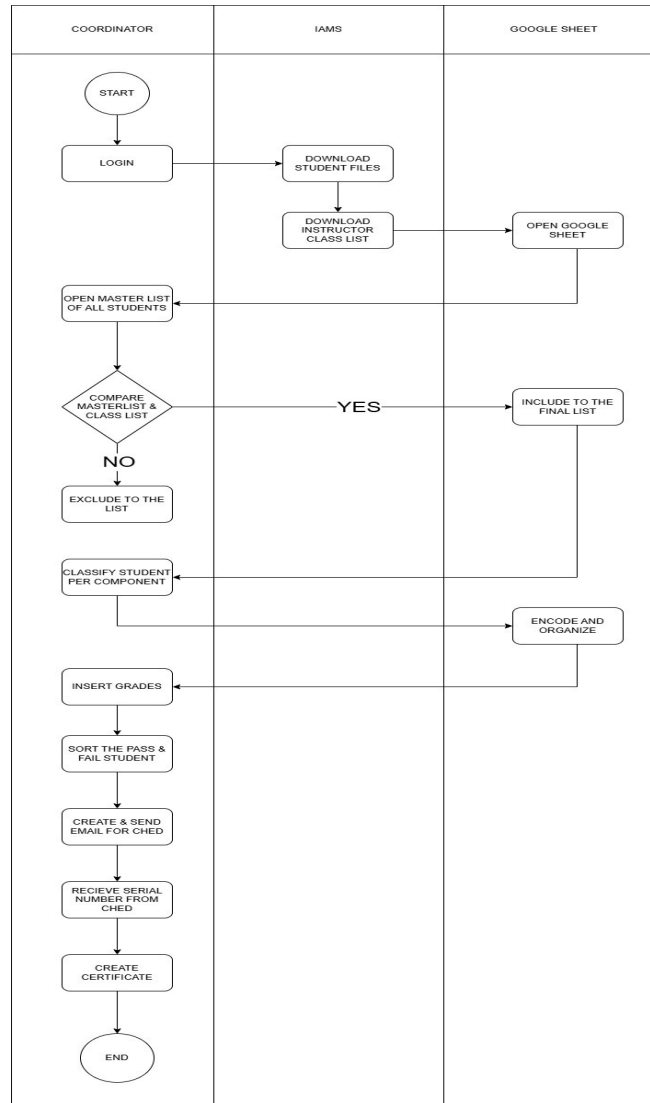


Figure 2: Existing Process of NSTP CWTS and LTS

Figure 3 shows the ROTC process of how students are assigned to ROTC Platoons in the manual process. Starting from the Coordinator logging into the system and downloading the list of student information from IAIMS, he or she gives the information to the ROTC Officer who sorts and assigns the students according to their capability and proficiency level to the respective platoons. The information is then inserted into the Google Sheets, where the respective platoons are finalized and NSTP officers are assigned to them. The Coordinator completes the process by reviewing the information for each brigade to confirm that the students are listed and assigned correctly.

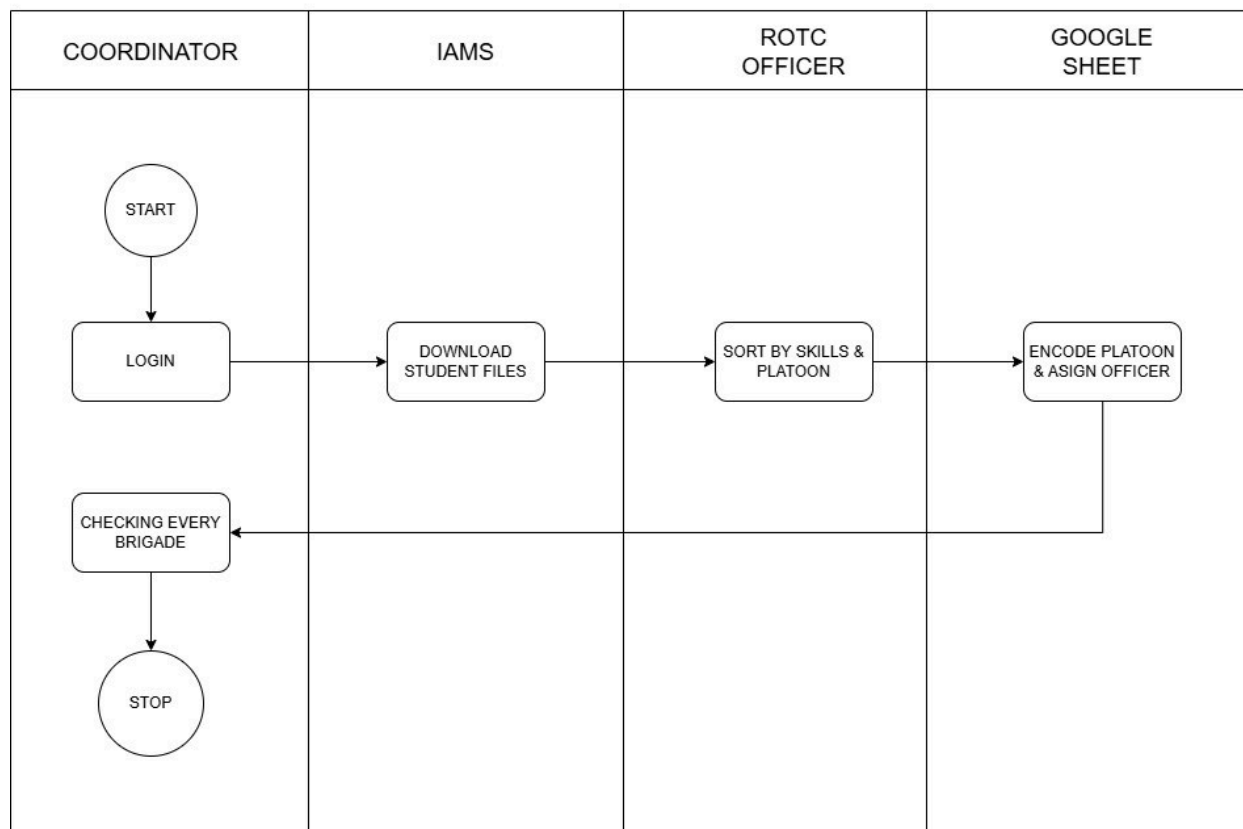


Figure 3: Manual Process for assigning students to ROTC Platoons

Figure 4 shows the process for creating and approving activities and reports. The ROTC Officer creates an approved activity design and accomplishment report that the Coordinator reviews for approval or rejection. The ROTC Officer is first required to create an activity design which the Coordinator reviews for approval or rejection. If the design is rejected, the ROTC Officer revises the design until it is approved. Then, the ROTC Officer creates the accomplishment report and submits it to the Coordinator for review. If there are errors in the report, it is revised again by the ROTC Officer. Once approved by the Coordinator, the completed report is archived, thus completing the process.

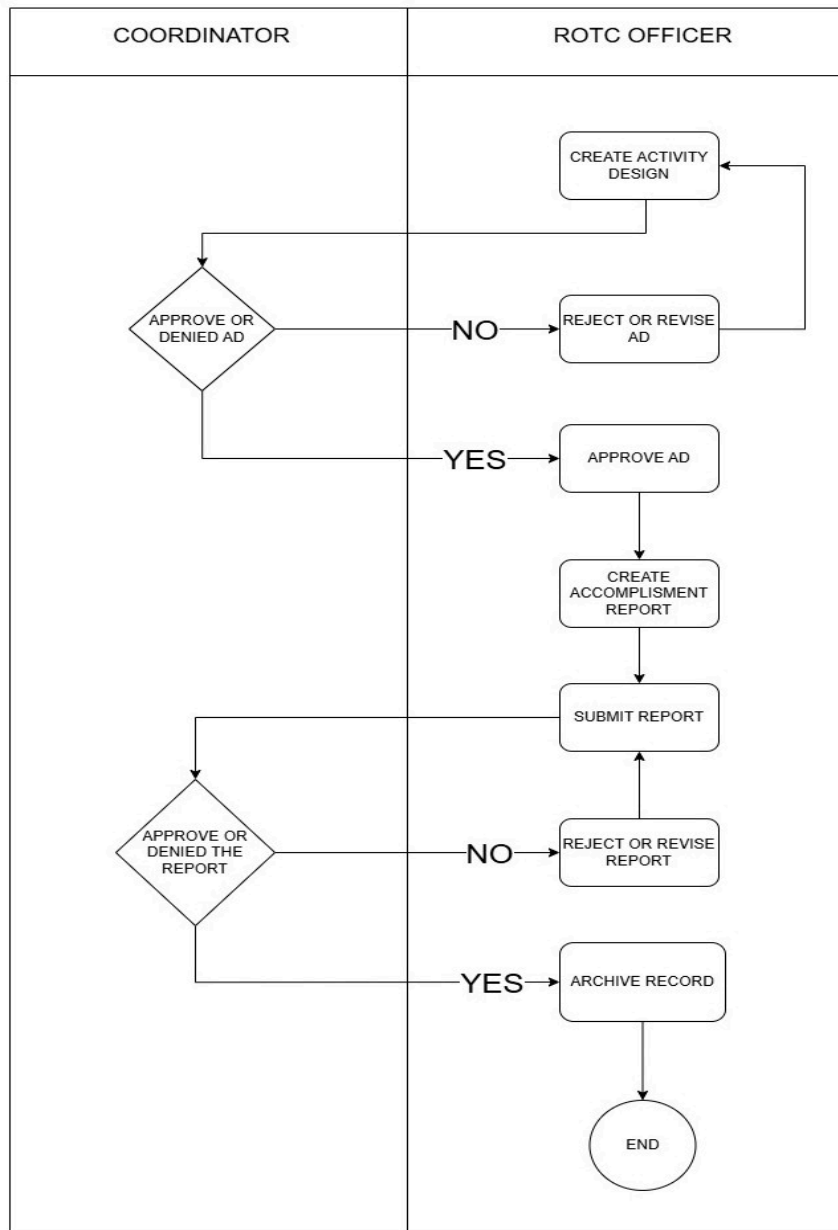


Figure 4: Existing Process of Activity Design and Report Approval

Figure 5 shows how the ROTC Officer begins by grading the students then after encoding and organizing in Google Sheets they sort them out to “Pass” or “Fail”. Grades are checked by the Coordinator if there are errors the ROTC Officer revises it. After correcting errors, the ROTC Officer emails the ROTC details to the admin. Then certificates are made once serial numbers are received.

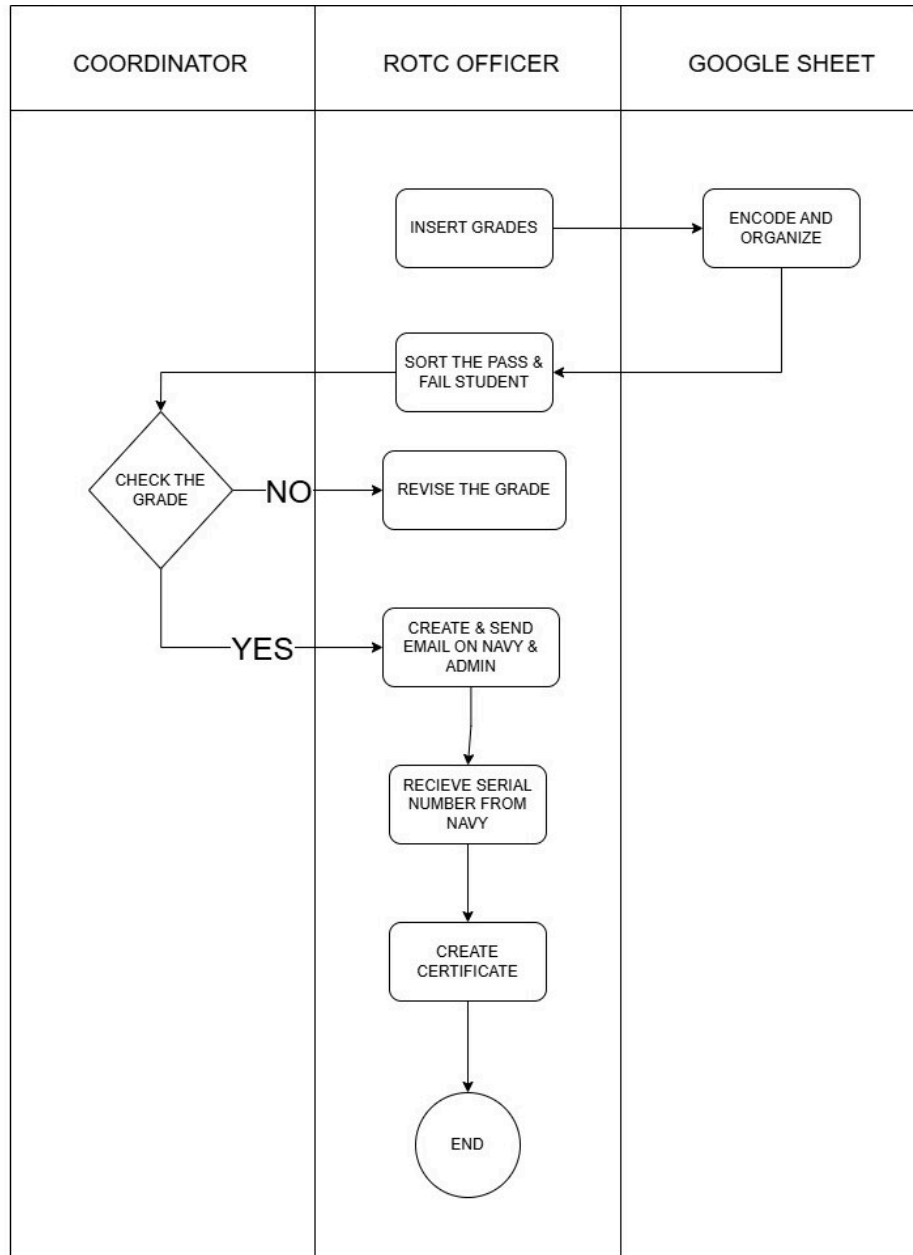


Figure 5: Existing Process of ROTC Grading and Certification

E. Objectives of the Project

a. General Objectives

The project primarily aims to design, develop, and evaluate a web-based Information Management System for the National Service Training Program (NSTP) office of Davao del

Norte State College that will centralize student records management, grading processes, improve report, and certificate generation, and enhance the operational efficiency in accordance with the ISO/IEC 25010 software quality standards.

b. Specific Objectives

Specifically, the project aims to:

1.Allow NSTP Coordinator to:

- 1.1 Log in to the system and manage account credentials (username and password)
- 1.2 Manage NSTP components such as CWTS, LTS, and ROTC
- 1.3 Assign instructors and organize NSTP sections per semester
- 1.4 Manage and update student records, including enrollment and NSTP assignments
- 1.5 Review and approve instructor-submitted activity plans and accomplishment reports
- 1.6 Upload official serial numbers provided by CHED for certificate processing
- 1.7 Monitor student completion status and NSTP compliance
- 1.8 Generate and print reports such as student performance and completion summaries
- 1.9 Manage system announcements and schedule NSTP activities through the calendar
- 1.10 View and monitor audit logs of system activities, including user actions such as logins, data updates, report submissions, and approvals for security and tracking purposes.

2.Allow NSTP Instructor/ ROTC 1st Class Officer to:

- 2.1 Log in and update account information.
- 2.2 View designated NSTP parts and student lists.
- 2.3 Create a list and assign platoons for ROTC.
- 2.4 Turn in activity plans and accomplishments reports.
- 2.5 Supervise report submission status of reports.
- 2.6 View announcements and scheduled activities of NSTP.

3. Allow the system to:

- 4.1 Store and record student, instructor, and NSTP data in a secure way.
- 4.2 Prohibit editing of reports after being endorsed by the NSTP Coordinator.
- 4.3 Process uploaded using OCR technology serial numbers of CHED and grades of students.
- 4.4 Automatically create NSTP completion certificates
- 4.5 Keep organized records by archiving data.
- 4.6 Develop summarized reports to monitor and evaluate.

F. Scope and Limitation

This project builds an information system that will facilitate the administration and implementation of the National Service Training Program (NSTP) through an online database and reporting system for efficient program implementation. The system has restricted user access for efficient program implementation. The NSTP Coordinator is the most active user, controlling user accounts, sections, and reports, and monitoring instructor activity logs. The coordinator can also monitor the activity logs of NSTP Instructors. Meanwhile, NSTP Instructors have limited control over the system. Instructors have limited control, can view activity reports, and enter grades for their assigned sections.

Hardware and software systems were required to make the ideas work. A trusted network would provide users with a far off cloud database 24/7. The 3D computer models were used to design physical objects, such as a desk. The program and processes in the system were planned out. The visual interface to the system was explored using Figma and Draw.io diagramming software, and coded in software applications such as Visual Studio Code. The final web interface of the system was used by users to generate and print high-quality certificates and reports on a printer.

The system maintains a robust database that tracks student and program information throughout the semester. It stores Student Profile Data, including ID numbers and contact information, as the primary identifier for alerts and communications. Additionally, the system stores Program Details for the individual components of CWTS, information pertaining to specific sections and assigned instructors for the LTS and ROTC track. Program Details, Training and Compliance Data, Grades, Community Project Data and other pertinent information are also stored in the system. After a student has gone through the program and has obtained passing grades, the system automatically creates a completion certificate using the serial numbers passed by CHED .

The proposed system begins with the Student Enrollment and Information Phase, where the students' information including their preferred NSTP component: CWTS, LTS, ROTC and the system captures student data for profiling. Academic Monitoring and Activity Phase, the coordinator assigns instructors, publishes schedules and announcements through the

Centralized Calendar, and monitors reports of each component through the online Dashboard. Additionally, the system integrates Optical Character Recognition (OCR), which begins when the coordinator uploads the grades from the IAIMS. The system automatically scans the file to extract the student names and grades, determining whether each student has passed or failed . In the separate process, the coordinator uploads the student information with their serial numbers then the system auto generates a certificate.

Although the system covers the main aspects of NSTP operations, it has certain limitations. It is only applicable to the management of NSTP within Davao del Norte State College and is not connected to other colleges or programs. The system is Internet-based, meaning users must have an Internet connection for the system to be fully functional; it is currently not mobile or offline. Moreover, the software is intended to be an aid, not an alternative, to administrative control. For academic integrity reasons, the NSTP Coordinator is still required to validate and approve critical data such as grades and student information.

G. Related Literature

The emergence of NSTP and student information systems have become a subject of research in tertiary education because of the necessity to efficiently manage student records and program activities. In a study conducted by Cajés and Durango [5], a web-based NSTP Management System was developed to respond to the challenges of manual record keeping, slow certificate processing, and ineffective monitoring of student activities. The researchers discovered that automation of NSTP processes enhances accuracy, efficiency, and data security in program administration. Similarly, Luciano, Nabong, and Manuel [6] developed a modern NSTP portal system to state universities and colleges in the Philippines to centralise the NSTP processes of student registration, monitoring, and reporting. Their analysis revealed that digital systems greatly decrease the manual workload and the overall management of the NSTP programs.

In higher education institutions, local studies of Montano and Almonte indicated that computerized student record management systems are more efficient than manual systems in storing, retrieving, and processing student data [7]. The results of their study further confirm the necessity to implement centralized systems in academic institutions to improve the accuracy and availability of data. The difficulties in managing student records were also mentioned by Riño [8] and include: data duplication, reporting delays, and poor record

management, thus necessitating the use of a Student Information Management System to enhance the administrative process. In addition, Grepon et al. [9] observed that centralized information systems are beneficial to educational institutions because they enhance data accuracy, accessibility, and security and reduce the workload of the administrators.

Also, with the assistance of Education Management Information Systems (EMIS), better decision-making becomes possible due to the provision of organized and reliable data [10]. In the Philippines, the emergence of digital transformation initiatives and systems such as HEMIS underscores the need to adopt technology to enhance institutional performance and delivery of services [11], [12]. The findings are used to develop the NSTP Management System in Davao del Norte State College to facilitate the operation and better control over the activities of the students.

The adoption of integrated information systems is also associated with increased efficiency and transparency within organizations. Research indicates that integration of systems can optimize workflow, minimize redundancy and improve productivity within institutions of learning [13], [14]. In general, local and international research evidences the necessity of the automated system that will enhance the processes within the institutional program like NSTP. All in all, they justify the necessity of an NSTP Information Management System that can enhance monitoring, record keeping, reporting and overall efficiency at Davao del Norte State College.

H. Related Systems

National service represents more than just providing public service. National service also promotes a form of nationalism and encourages citizens. Some countries have established forms of national service that may either take a military format or a nonmilitary format; additionally many countries have established national service programs associated with education. An example of an educational based program would be the National Service Training Program (NSTP) in the Philippines.[1] "The system titled 'Intellischool: A Student Information System for Senior High School' provides the means to develop a digital student database and management system." This digital system manages student information efficiently throughout school systems. The "system" automates processes for students such as student records management, enrollment process management, and information tracking. Therefore, it eliminates manual labor and improves both accuracy and access to student information.

The implementation of automation for the management of the Davao Region (Region XI) has shown its commitment to this approach. Automated student management systems are created to integrate key aspects of the educational institution's function; such as enrollment, student records, scheduling, and reporting into one or more central systems. An example of this can be seen at the University of Southeastern Philippines where they have an Online Enrollment System (OES), Student Records Information System (SRIS) and U.S.E.P. Virtual Learning Environment (UVE). These three systems provide the means for students to manage their own transactions for enrollment, provide students with online access to educational resources and enable them to monitor and maintain control over their own academic records through various web-based applications. This model also provides support for digital based learning and centrally managed access to students' information through their own accounts. Therefore, it demonstrates a similar application to what will be required from the proposed NSTP Management System for providing centralized access to enrollments, tracking academic records, processing instructors' reports and other functions as part of an overall integrated student management solution [15] .

Silliman University introduced an online NSTP registration and hybrid activity management process, where students registered online, received a digital notification, and participated in online and in-person NSTP activities. This setup illustrates how an institution can integrate online registration, schedule and announcement functions in managing NSTP operations, which is similar to the enrollment, calendar, and announcement functions of the proposed system [16]. Similarly, University of the Philippines Diliman embraced on-line NSTP coordination in the form of webinar-based common modules, online tracking of graduates, and centralized NSTP administrative planning. Its application emphasizes the application of digital coordination tools to large-scale NSTP operations, particularly in scheduling, reporting, and overall monitoring of the program [17].

Another system of Web based File management system office of national service training program in the university of Sorsogon State University and is aimed at improving the inefficient manual record handling in the NSTP offices by developing a web based system that centralizes the data storage and automates file management. The system allows student records to be accessed more quickly, organized in a better way, and handled more accurately, improving the overall efficiency of administration. This is linked to the proposed system where both objectives are to digitalize and simplify the processes of NSTP but the proposed system can consist of other features like user role management, and generation of reports [18].

The system Inspirational Solutions-Design and Implementation of an Advance National Service Training Program (NSTP) Portal to State Universities and Colleges in the Philippines focuses on introduction of an electronic centralized portal to improve the management of NSTP in the institutions of higher learning. The system integrates the required functions to ensure that student information management, activity monitoring and communication between administrators, instructors and students are integrated to resolve the typical issues that include fragmented data and inefficient manual processes. The paper explains how a web-based portal can be applied to enhance access, stream operations, and overall coordinate the program [19].

Existing systems mainly focus on general student data management but do not fully support NSTP-specific processes because of such limitations, there is a necessity to design a specific NSTP Information Management System to be able to service such specialized needs and enhance the overall program efficiency. Such a system would make it possible to organize the NSTP records in a more optimal way, simplify the communication between administrators, instructors, and students, and automate the most important processes to make the work of administrators, instructors, and students easier . In addition, it can also be utilized in order to enhance monitoring and reporting of the activities of the NSTP, timely development of documents, and provision of a central platform through which one can easily make decisions and manage the whole program.

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Chapter 1

INTRODUCTION

A. Background of the Study

National service is more than just "serving the public", it involves serving under specific political and economic systems, promoting nationalism, and helping to build that nationalism. Various countries around the world implement national service programs, these programs can be military, non-military, or compulsory, and some are linked to the educational system, such as the National Service Training Program (NSTP) in the Philippines [1]. Research has shown various impacts of service learning on students in different countries. These studies usually show that service learning has an impact on students in ways

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CHAPTER II

METHODOLOGY

This chapter carefully organizes the project by considering the requirements, implementation timeline, and feasibility study. The methodologies include different techniques such as surveys, interviews, observations, and other methods used to collect relevant data. Those methodologies provide a solid foundation for the system, ensuring accurate results and improving the overall validity and credibility of the project.

A. Project Initiation

a. Software Development Life Cycle

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b. Work plan and Schedule Feasibility

Project Milestones and Deliverables

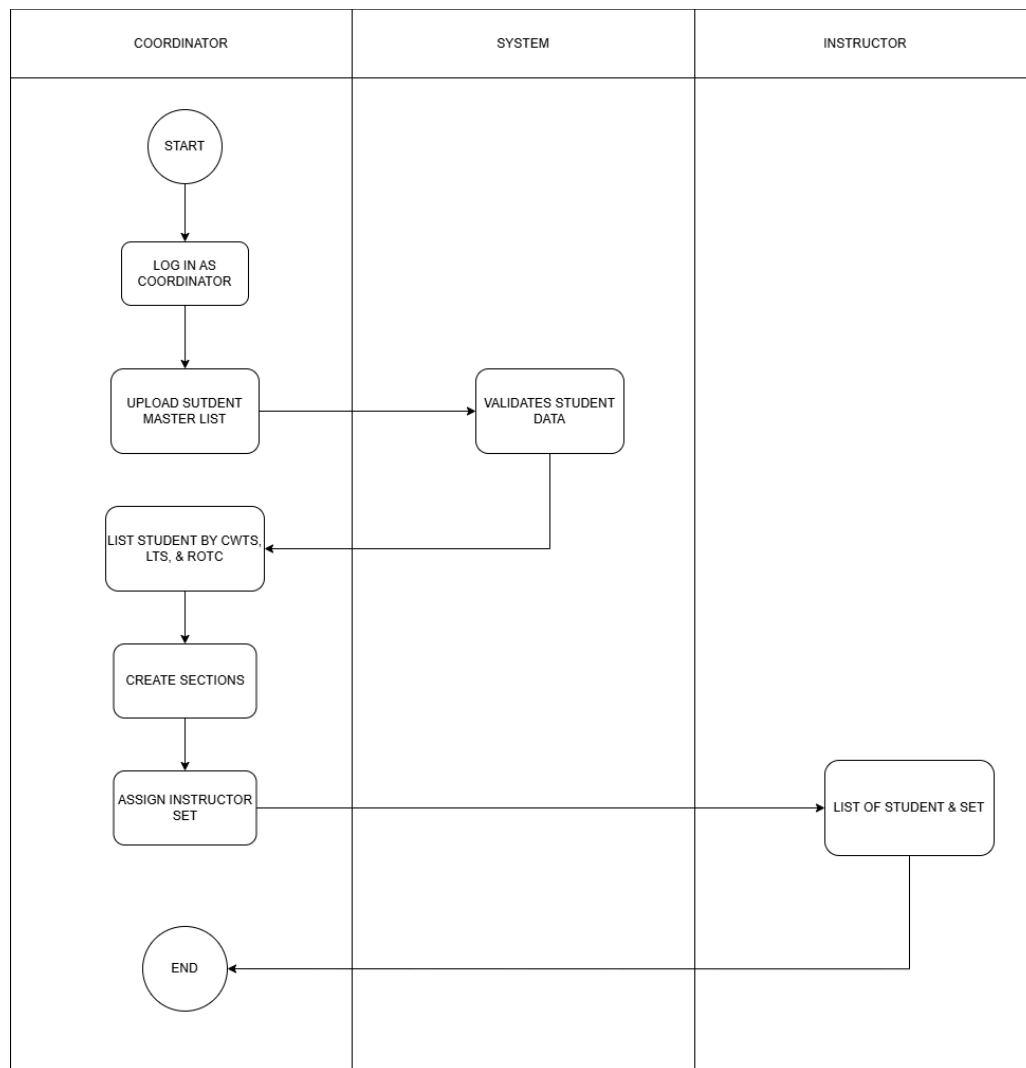
The NSTP Management System project will follow a systematic and iterative process to ensure that it is completed on time and effective for Davao del Norte State College. The process begins with interviews to determine how student records are currently being managed by the NSTP coordinators and staff and what problems they encounter in manually tracking student records. The team shall then design the layout and database of the system using Draw.io, php and decide the look of the faculty and student portal.

After the design is complete, the team will start coding the site using Visual Studio Code and XAMPP, with the student record management module. A major effort will be on developing the OCR technology to enable the system to read and process student grades and CHED serial numbers automatically to generate certificates for students. All features, including automated certificate generation and audit log tracking, will be tested for errors to make sure everything is as high quality as possible. The team will also solicit opinions from the NSTP staff on how to make the dashboard easy to navigate. The Coordinator and Instructors will be trained prior to the final launch with the ability to handle reports and make announcements. The project will be completed in a timely manner and be ready for use at the beginning of the next semester, and will be aligned with the current skills of the team..

Integrated and Proposed New Processes

The proposed new process for the National Service Training Office of Davao del Norte State College has been made to make things easier and faster for both the coordinator, instructors and 1st class officers. Instead of doing things manually or using separate tools, everything is now connected in one system. The admin can just open a web browser and access the NSTP Management System directly to manage records. Users will use a web server that has both frontend and backend to do their tasks, All data is saved in one database server, so it's easier to find and update. Files are also stored properly in a file repository, with a backup just in case. This setup avoids double work and saves time because everything is more organized. Overall, the system helps the office work more smoothly.

Figure 2, shows the CWTS and LTS Master List Process of the Coordinator logs in to the platform to add the raw student master list, beginning the CWTS & LTS Masterlist Process. Once uploaded, the data automatically checks for its correctness and structure. Once the workflow is successfully validated, the Coordinator will receive a classified list of students by their particular programs (CWTS and LTS). This ordered data is used to assign a specific instructor to each group of official class sections by the Coordinator. This will essentially finish the process by notifying the teacher and enabling him/her to see the list of students assigned to him/her and the timeline for which the teacher made the class.



F

Figure 2. Proposed CWTS and LTS Master List Process

Figure 3, shows the CWTS and LTS Activity Process initiates with the Coordinator logging in to manage the calendar, prompting the System to automatically publish these calendar updates. Next, the Coordinator establishes the activity plan guidelines and sets a specific submission deadline. This action notifies the Instructor to create and

submit their proposed activity plan back through the System. The Coordinator then reviews the submitted plan; if it is lacking, it is returned to the Instructor for revision or decline, but if it meets expectations, it is officially approved. Following this approval, the Coordinator sets a new deadline for the final accomplishment report. The Instructor then drafts and submits this accomplishment report via the System. The Coordinator conducts a final review, either returning the report for further revisions if necessary, or approving it. The workflow concludes with the Coordinator continuously monitoring the overall status of the instructor's submissions until the end of the activity cycle.

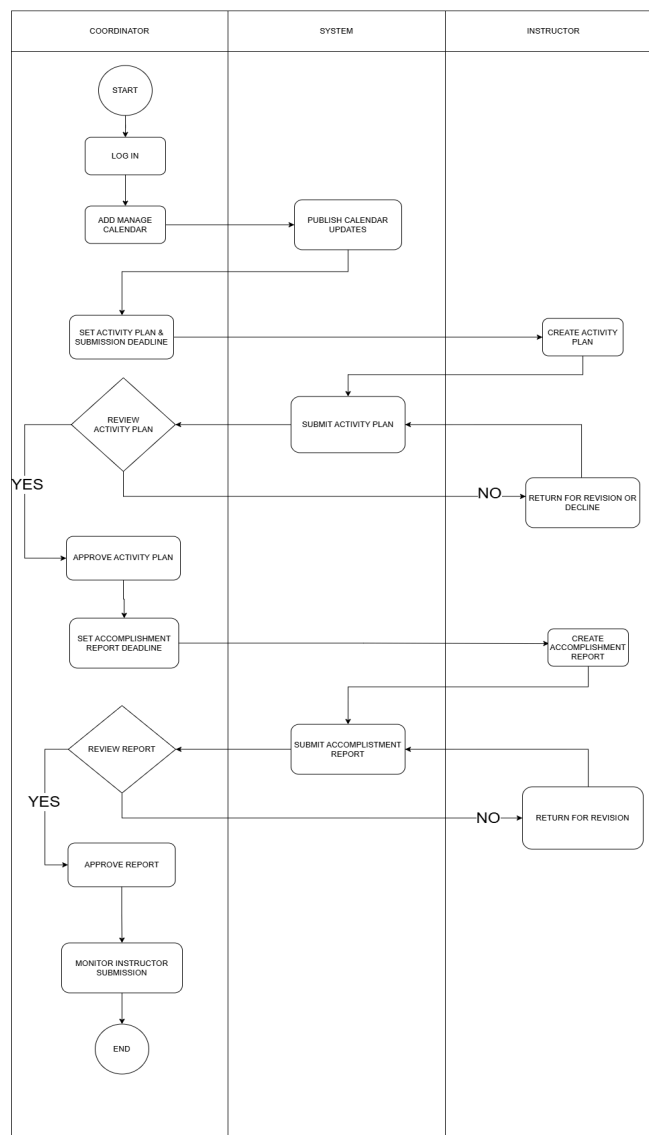


Figure 3. Proposed CWTS and LTS Activity Process

The Coordinator uploads grades of students into the platform and the CWTS and LTS Grades Process starts from there as shown in Figure 4. The System then automatically evaluates the data uploaded to identify whether each student has passed or failed. When a student passes, it automatically pulls and saves his/her grades. If a student fails, the record automatically archives and is not included in the passing list, except for the Coordinator to exclude the record from the passing list if they choose. After this assessment the Coordinator ensures and confirms the overall grade. After being finalized, the workflow goes to the System to create official certificates and record the students' completion status. This process ends when the Coordinator updates the audit logs, reviews the reports and analytics and finally archives records.

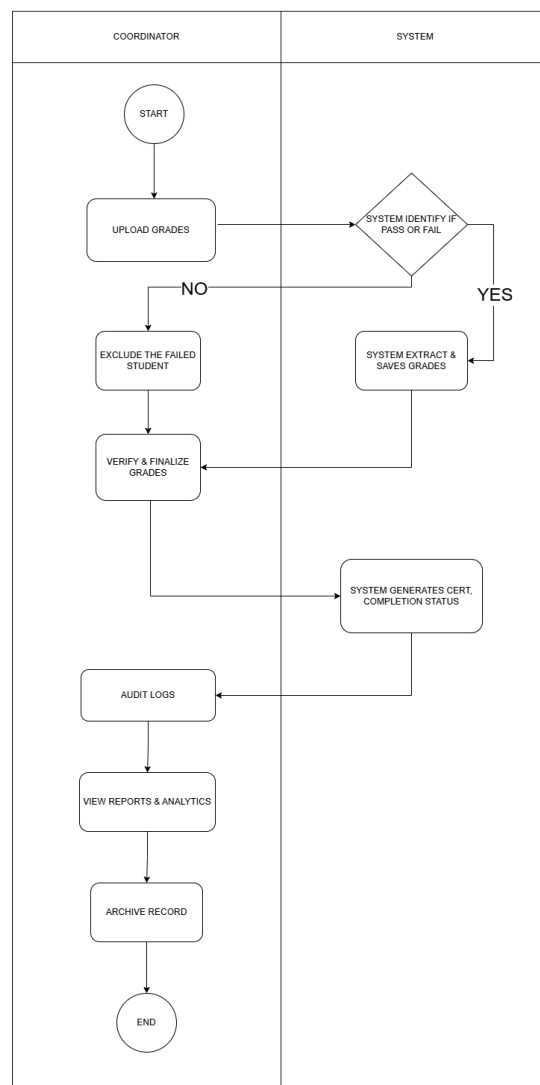


Figure 4. Proposed CWTS and LTS Grades Process

Figure 5, The ROTC Masterlist Process begins with the Coordinator logging into the platform to upload the initial student master list. Once uploaded, the System receives and securely stores this raw student data. The workflow then shifts to the ROTC Officer, who accesses the stored list to organize the students by assigning them to specific platoons and specialties. Following this categorization, the ROTC Officer creates the official sections and assigns a specific officer to oversee each set. These finalized section assignments are then submitted back to the System. The process concludes when the Coordinator receives the organized roster, detailing the complete list of students alongside their newly assigned officers in charge.

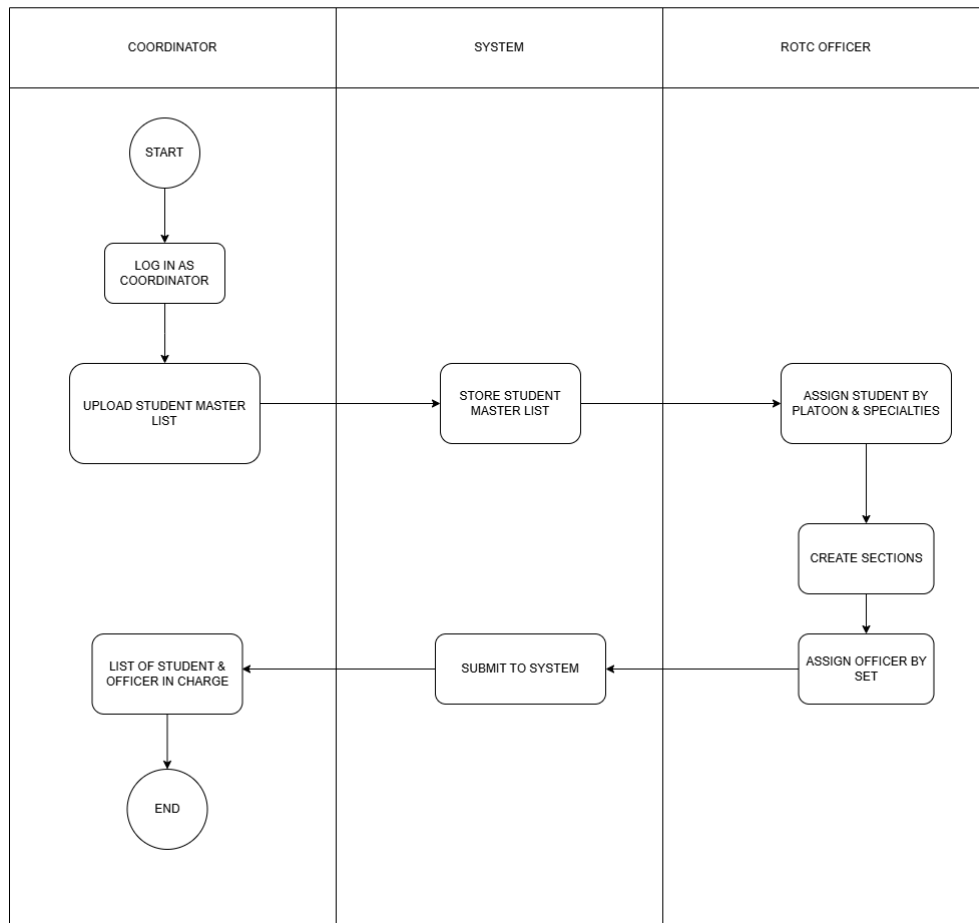


Figure 5. Proposed ROTC Master List Process

Figure 6 shows the ROTC Activity Process begins with the Coordinator logging into the platform to manage the calendar, which prompts the System to publish the official calendar updates. Next, the Coordinator establishes the activity plan guidelines and sets a submission deadline. The workflow then shifts to the ROTC Officer, who creates the required activity plan and submits it back through the System. The Coordinator then reviews this submitted plan; if it is not approved, it is returned to the ROTC Officer for revision or decline. Once the plan is approved, the Coordinator sets a new deadline for the final accomplishment report. Following this, the ROTC Officer creates and submits the accomplishment report via the System. The Coordinator reviews this final report. If it is rejected, it is sent back to the officer for further revision. Upon approval, the Coordinator monitors the overall officer submission compliance, successfully concluding the process.

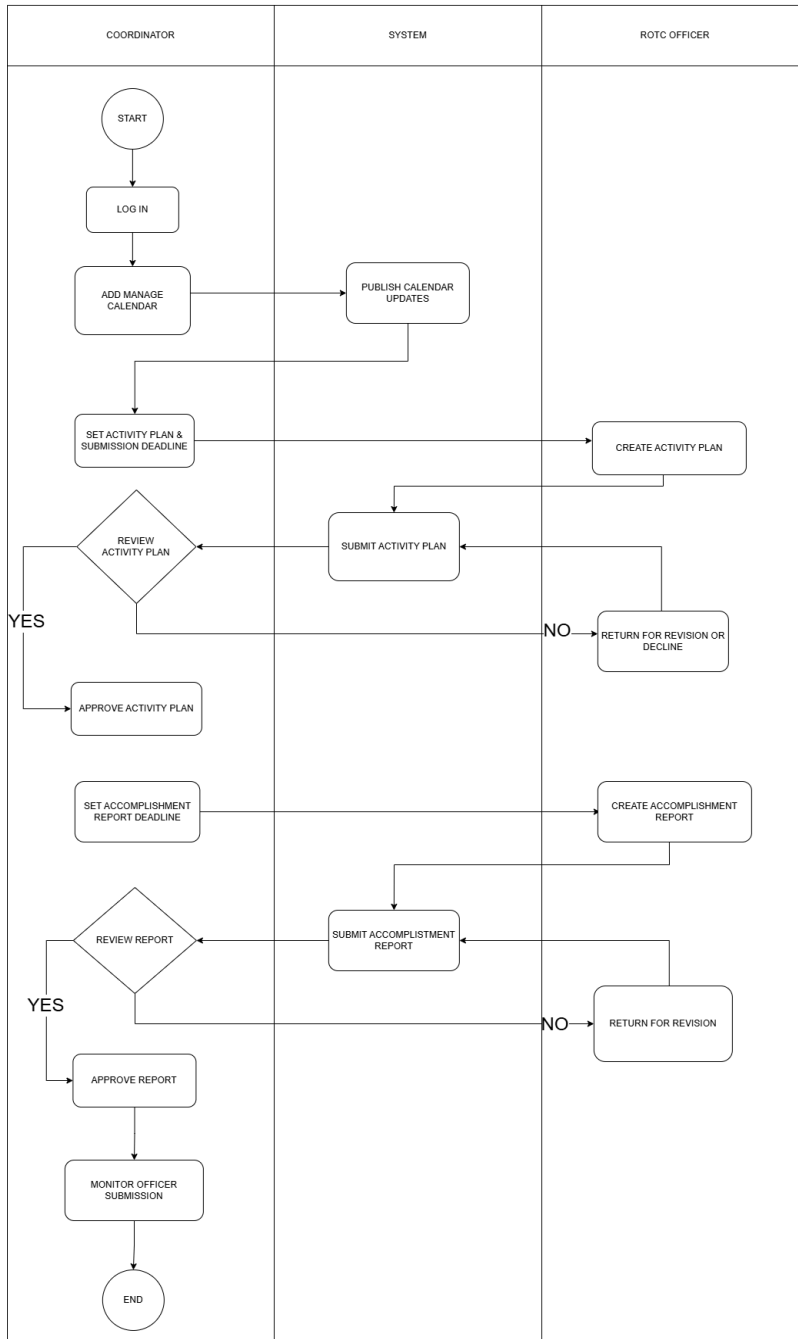


Figure 6. Proposed ROTC Activity Process

Figure 7, shows the ROTC Grades Process starts with the ROTC Officer logging into the platform to upload the students' grades. The System then automatically evaluates this uploaded data to identify whether each student has passed or failed. If a student fails, the workflow returns to the ROTC Officer, who is tasked with excluding the

failed student from the passing list. Conversely, if a student passes, the System automatically extracts and saves their grades. Following this, the Coordinator steps in to verify and finalize the official grades. This finalization prompts the System to generate the students' certificates and update their completion statuses. The process successfully wraps up with the Coordinator managing the audit logs, viewing the generated reports and analytics, and finally archiving the records.

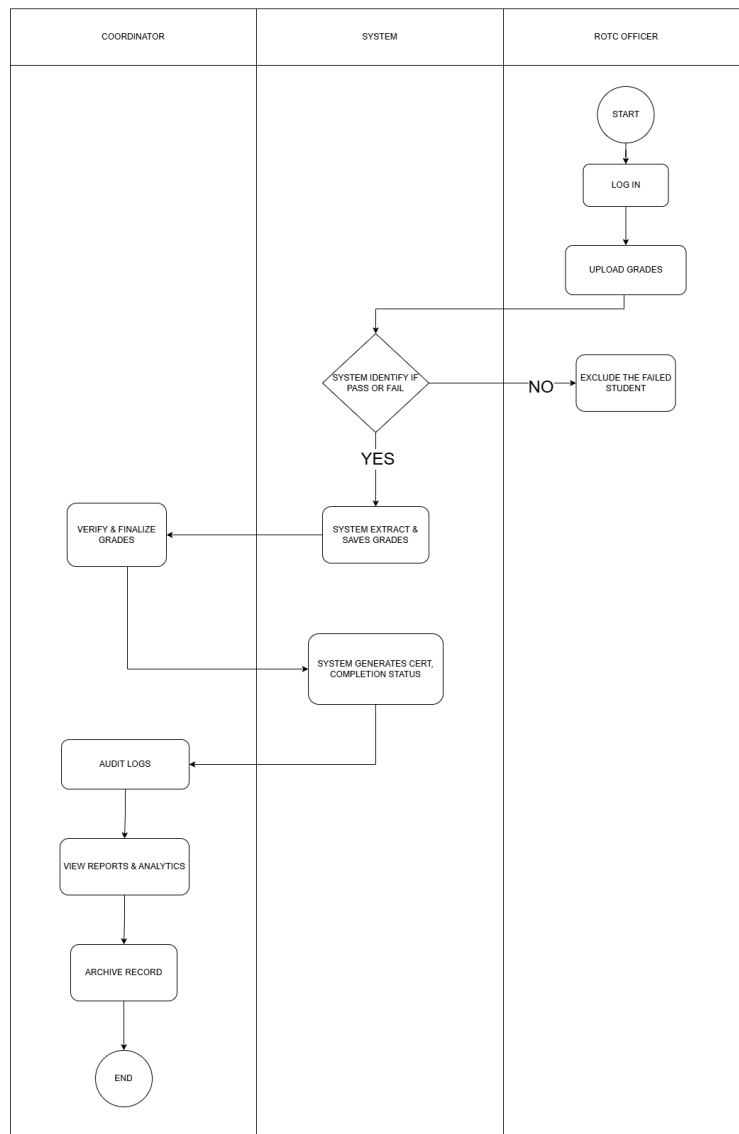


Figure 7. Proposed ROTC Grades Process

Development Team

A development team works with supportive qualities. It is generally agreed that the person spearheading the project should understand project management knowledge [1]. The project manager directs and supervises the team's work while ensuring that schedules and budgets are met. System developers translate designs into functional code. System analysts operate as the link between business and technology, analyzing needs and translating them into executable specifications. System designers create the system's blueprint, which defines its structure and functionality. This collaborative approach promotes a successful development process, resulting in projects that meet both business and technical requirements.

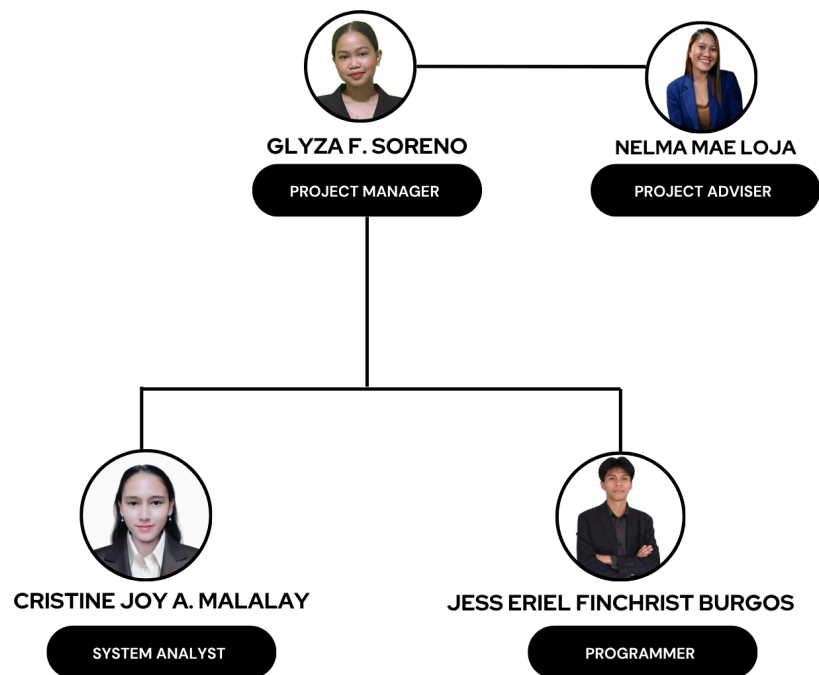


Figure 12: Project Team Organizational Structure

c. Feasibility

Operational Feasibility

The proposed National Service Training Program (NSTP) Management System is highly feasible to operate within Davao del Norte State College. Currently, the NSTP office faces problems like manual encoding, lost records, and slow processing of certificates. This new system will solve these issues by placing all student records, grades, and activities into one secure online platform.

The main users are the NSTP coordinator, ROTC coordinator, instructors, and student officers will find the system easy to use because it changes heavy manual work into automated tasks. The school already has the basic computers and internet connection needed to run this web-based system. With brief training, the staff will be able to easily adopt this new process and improve their daily work speed and accuracy.

Technical Feasibility

The implementation of the National Service Training Program (NSTP) Management System at the Office of Davao del Norte State College in Panabo City is technically feasible because all the tools and equipment needed are already available and easy to manage. The system will be built using Visual Studio Code for coding, MySQL for the database, and XAMPP to run everything so NSTP staff can use it on their current computers- no need to buy new ones.

Table X. Business's Existing Technology Template

Technology	Particulars and Specification	Quantity
Software		
Visual Studio Code	Version 1.80+ (Open Source Code Editor)	1 unit
Xampp	v8.2.4 (PHP)	1 unit
Draw.io	Online/Desktop Version (Diagramming Tool)	1 unit
Web Hosting (yearly)	Yearly Subscription (Linux-based with	1 unit

	MySQL Support)	
Hardware		
Laptop	Intel Core i5/i7, 8GB-16GB RAM, 512GB SSD	3 units
Printer	All-in-One Inkjet/Laserjet (Print/Scan/Copy)	1 unit
Router	Dual-band Wireless AC/AX Router	1 unit
Network		
Fiber Internet	High-speed Fiber Optic Connection (100Mbps+)	1 subscription
Local Area Network	Cat6 Ethernet Cables and Campus Wi-Fi Mesh	1 system
Network Security	SSL Certificate (for HTTPS data encryption)	1 unit

d. Stakeholder Analysis

Stakeholder analysis is the process of finding the people or groups who are affected and can affect the project. This helps the team know how to work with, what each person needs, and how to keep them involved throughout the development of the NSTP Management System..

Table X. Stakeholder Analysis Template

Stakeholder name	Impact	Influence	What is important to the stakeholder ?	How could the stakeholders contribute to the project?	How does the stakeholder block the project?	Strategy for engaging the stakeholders?
NSTP and ROTC Coordinators	High	High	It is important to have a centralized	The coordinators can guide the	If the coordinators do not approve the	Present a clear project proposal showing how the system will

			system instead of doing things manually. An easy way to manage student lists, automatically check pass/fail grades, and quickly create certificates that meet the school's academic standards.	project by giving approvals, and help promote the system to other departments.	project and are not aligned with school priorities, the development may be delayed or stopped. And can resist change if they are used to the old manual process	improve the NSTP and benefit the school. Give regular updates on the project progress and make sure the system follows school policies.
NSTP Instructors and 1st Class Officers	High	Medium	A simple and easy system to view class lists, submit activity plans, accomplishment reports, and check grades without confusion.	Instructors and 1st class officers can contribute by helping test the system, give feedback, provide actual activity and grade data that will be managed by the system	If instructors and officers refuse to use the system, the system will not work well. Lack of cooperation in submitting reports through the system can cause data gaps	Involve in the early design process so the system fits their actual needs.

e. Project Charter

The Project Charter serves as the official authorization for the development and implementation of the National Service Training Program (NSTP) Management System for Davao del Norte State College. It outlines the project's scope, core objectives, and the key stakeholders involved. This document acts as a shared agreement between the development team and the college administration, ensuring that all parties are aligned with the goal of digitizing student records and automating certificate generation. By defining the roles, responsibilities, and success criteria at the start, this charter provides the formal authority needed to utilize college resources and ensures that the system is developed to meet the specific administrative needs of the NSTP office.

Table X. Project Template

National Service Training Program (NSTP) Management System for Davao del Norte State College	
Project Description	To centralize student records, automate the creation of certificates using CHED serial numbers, and make the monitoring of NSTP reports faster and easier.
Project Objectives	a. General Objectives The project primarily aims to design, develop, and evaluate a web-based Information Management System for the National Service Training Program (NSTP) office of Davao del Norte State College that will centralize student records management, grading processes, improve report, and certificate generation, and enhance the operational efficiency in accordance with the ISO/IEC 25010 software quality standards. b. Specific Objectives Specifically, the project aims to: 1.Allow NSTP Coordinator to: 1.1 Log in to the system and manage account credentials (username and password)

	1.2 Manage NSTP components such as CWTS, LTS, and ROTC 1.3 Assign instructors and organize NSTP sections per semester 1.4 Manage and update student records, including enrollment and NSTP assignments 1.5 Review and approve instructor-submitted activity plans and accomplishment reports 1.6 Upload official serial numbers provided by CHED for certificate processing 1.7 Monitor student completion status and NSTP compliance 1.8 Generate and print reports such as student performance and completion summaries 1.9 Manage system announcements and schedule NSTP activities through the calendar 1.10 View and monitor audit logs of system activities, including user actions such as logins, data updates, report submissions, and approvals for security and tracking purposes. 2.Allow NSTP Instructor/ ROTC 1st Class Officer to: 2.1 Log in and update account information. 2.2 View designated NSTP parts and student lists. 2.3 Create a list and assign platoons for ROTC. 2.4 Turn in activity plans and accomplishments reports. 2.5 Supervise report submission status of reports. 2.6 View announcements and scheduled activities of NSTP. 3. Allow the system to: 4.1 Store and record student, instructor, and NSTP data in a secure way. 4.2 Prohibit editing of reports after being endorsed by the NSTP Coordinator. 4.3 Process uploaded using OCR technology serial numbers of CHED and grades of students. 4.4 Automatically create NSTP completion certificates 4.5 Keep organized records by archiving data. 4.6 Develop summarized reports to monitor and evaluate.
The project is considered successful when:	The project is considered successful when the NSTP Management System is fully implemented and used effectively by NSTP staff.

Title	NATIONAL SERVICE TRAINING PROGRAM (NSTP) MANAGEMENT SYSTEM FOR DAVAO DEL NORTE STATE COLLEGE
Project Sponsor:	Razul A. Atowan Mark Anthony G. Librea
	Project Manager: Glyza F. Soreno
	Programmer: Jess Eriel Finchrist M. Burgos
	System Analyst: Cristine Joy A. Malalay
Available Resources	Microsoft Windows 11
Milestone	Status
	Planning, Execution, Development, and Project Closure
Potential Risk	
Approval	NATIONAL SERVICE TRAINING PROGRAM (NSTP) MANAGEMENT SYSTEM FOR DAVAO DEL NORTE STATE COLLEGE
	Sponsor: Razul A. Atowan Signature: _____ Sponsor: Mark Anthony G. Librea Signature: _____ Project Manager: Glyza F. Soreno Signature: _____

f. Project Team Roles and Responsibilities

The project is managed by a dedicated team with clear responsibilities to ensure its successful development, testing, and delivery. Each member is in charge of a crucial aspect of the development process, collaborating to turn the DNSC NSTP Office's needs into a workable digital solution. The team utilizes a collaborative approach that has the design, development and quality testing going on at the same time for efficiency. The team can swiftly find technical problems and guarantee that the finished product is safe, easy to use, and able to manage the large amount of student data needed by the college by utilizing a shared development environment.

Team Member	Roles	Responsibility
Glyza F. Soreno	Project Manager	Provides leadership and serves as the main point of contact with the NSTP Office. She keeps track of the project timeline, helps resource the project, and makes sure all milestones are met on time. She also has the responsibility for final quality checks and the project's documentation.
Cristine Joy A.Malalay	System Analyst	Responsible for creating technical blueprints from user needs. She designs the database structure and how the system will work with it using Draw.io. She makes sure the system logic is correct for ROTC platoons, CWTS/LTS components and OCR processing requirement
Jess Erielfinchrist M. Burgos	Programmer	Programmer Technical construction of the system in Visual Studio Code and PHP/MySQL. He creates the secure login systems, installs the audit logs, and deploys to the web server and makes sure the network is stable for the school-wide use.

Table X. Roles and Responsibility Template

a. Work Breakdown Structure

The Work Breakdown Structure (WBS) divides the entire project into smaller, easier-to-manage parts. Each part represents a specific task or phase in building the NSTP Management System. This helps the team track progress, assign responsibilities, and manage the timeline properly.

Figure X shows how the project is organized from start to finish. The main project is broken down into five major phases: Project Initiation, requirements and design, system development, testing and evaluation, and deployment and closeout. Each phase is then divided into smaller tasks that the team needs to complete.

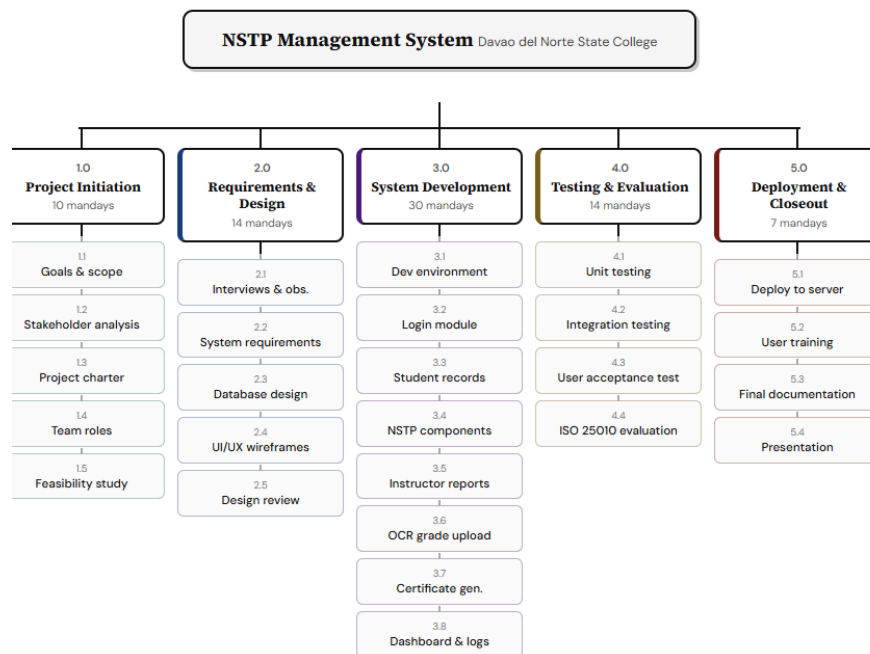


Figure x. WBS

b. WBS Dictionary

The WBS Dictionary is a detailed document that describes the technical and narrative requirements for each work in the NSTP Management System. Table X shows the task titles to give an overview of what the tasks are, but the dictionary is there to guarantee that there is no ambiguity in determining when a task is completed; it specifies precisely what tasks, deadlines, and deliverables are required to close a task. A simple instrument to assist the project team to be familiar with the quality needs and the contribution of each stage. With the dictionary, you can accurately keep track of projects and keep important data such as Man-days, Schedules, Actual Costs, etc. Provides the project manager with a way to monitor the progress of the system development process and ensure that the project remains on track to complete within the project's scope and budget from requirements collection through project closeout.

Table X. Work Breakdown Dictionary

WBS Code	Phases and Activity	Duration (Mandays)	Star Date	End Date	Actual Cost
1.0	Project Initiation	10	Feb-02-2026	Feb-13-2026	
1.1	Goals and scope	2	Feb-03-2026	Feb-03-2026	
1.2	Stakeholder analysis	2	Feb-04-2026	Feb-05-2026	
1.3	Project charter	3	Feb-06-2026	Feb-09-2026	
1.4	Team roles	2	Feb-010-2026	Feb-11-2026	
1.5	Feasibility study	2	Feb-12-2026	Feb-13-2026	
2.0	Requirements and Design	14	Feb-16-2026	March-05-2026	
2.1	Interviews and observations	3	Feb-16-2026	Feb-18-2026	
2.2	System requirements	3	Feb-19-2026	Feb-23-2026	
2.3	Database design	3	Feb-24-2026	Feb-26-2026	
2.4	UI/UX wireframes	3	Feb-27-2026	Mar-03-2026	
2.5	Design review	2	Mar-04-2026	Mar-05-2026	
3.0	System Development	78	Mar-06-2026	May-30-2026	

h. Gantt Chart

This part shows the NSTP Management System's visual development plan. It provides a comprehensive framework that outlines the project in various phases: starting at the NSTP Office and ending with staff training and deployment. They can then follow each task's progress in real time on this chart to ensure that key capabilities, such as the OCR integration and ROTC platoon management, are completed in a timely manner. This timetable has been carefully planned to coincide with the academic cycle of the college. It ensures that the system shall be fully trialled and tested, secure and in operation before the start of the next semester. Moreover, the chart helps the project manager organize the programmer and the system analyst's tasks, so that the programmer can coordinate to work with the system analyst, and otherwise the two members can manage their time appropriately and handle any technical issues that may arise at the early stages of development.

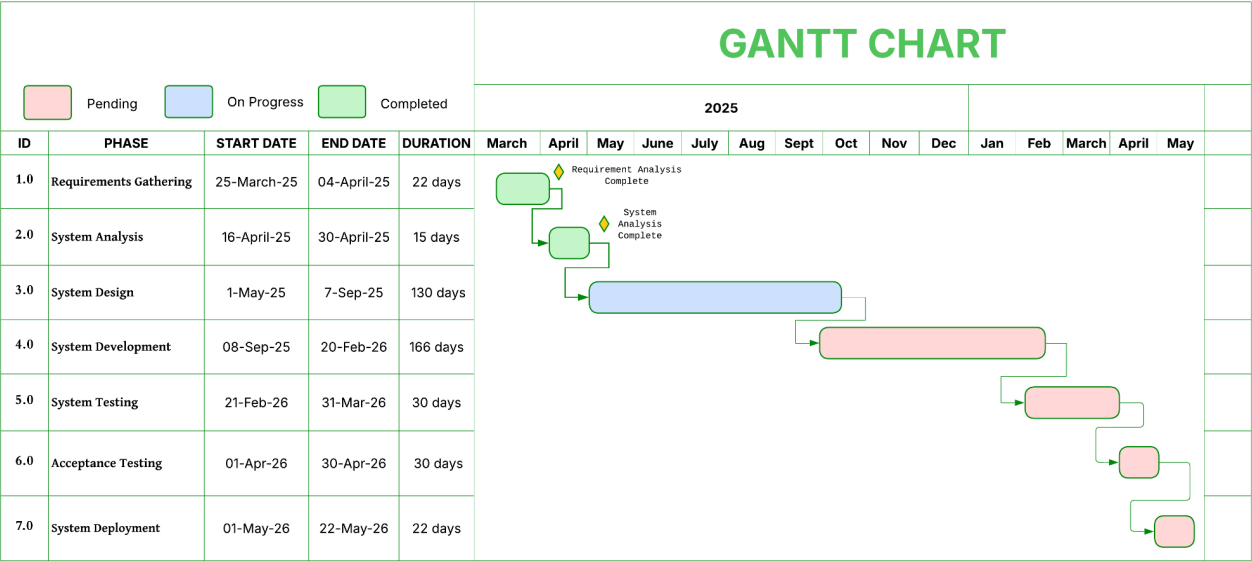


Table X: Gantt Chart

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Table X. Budget Summary of the Capstone Project

Phase	Duration (Man-days)	Start Date	End Date	Actual Cost
Project Initiation	10			
Requirements and Design	14			
System Development	78			
Testing and Evaluation				
Deployment and closeout				
Total Man-days		Total Actual Cost		

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Table X. Manpower Cost Estimates

Position	Man-days	Rate/Day	Actual Cost
Project Manager	25	250.00	
Project Developer	30	250.00	
Project Analyst	20	250.00	
TOTAL			

b. Budget Allocation and Schedule

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Table X. Budget Allocation per Month

YEAR	MONTH	WBS COST	Technology Cost	Audit Cost	Effort Cost	Total Cost	Cumulative Cost
2026	February						
	March						
	April						
	May						
	June						
	July						
	August						
	September						
	October						
	November						
	December						
2027	January						
	February						
	March						
	May						
Total							

Effort Cost

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Table 10. Effort Cost per Resource

Role	Communication	Meeting	Reporting	Actual Cost
Project Manager				

Project Developer				
Project Analyst				
Total				

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Cash Outlay

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Table X. Hardware: Actual Cost

Hardware	Quantity	Unit Price	Total Cost
TOTAL			

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Table X. Software: Actual Cost

Software	Quantity	Actual Cost
Total Cost		

c. Financial Viability

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d. Cost and Benefit Description

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Table X. Time Spent: Current vs. Proposed Process

Processes	Current	Proposed	Difference
Total			

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Table X. Expenses: Current Process of Facility Manager

Processes	Facility Manager (₱76.25/hour)	TOTAL
Total		

(Add more tables based on your users)

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e. RACI Chart

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Table X. RACI TABLE

Legends						
Responsible	Assigned to complete the given task					
Accountable	Has the final decision-making and accountability. Only one person per task.					
Consulted	An adviser and shareholder in decision-making.					
Informed	Must be informed about the decision and action.					
Rental Management System for Davao del Norte State College (DNSC)						
Main Tasks and Milestones	Project Manager	System Analyst	System Programmer	System Designer	User Tester	Project Sponsor
Gather system requirements	A	R	C		C	I
Design database and website	A	R	C		I	I
Code the web and ocr	A	C	R		I	I
Conduct system testing	I	A	C		R	C

Functional Requirements:

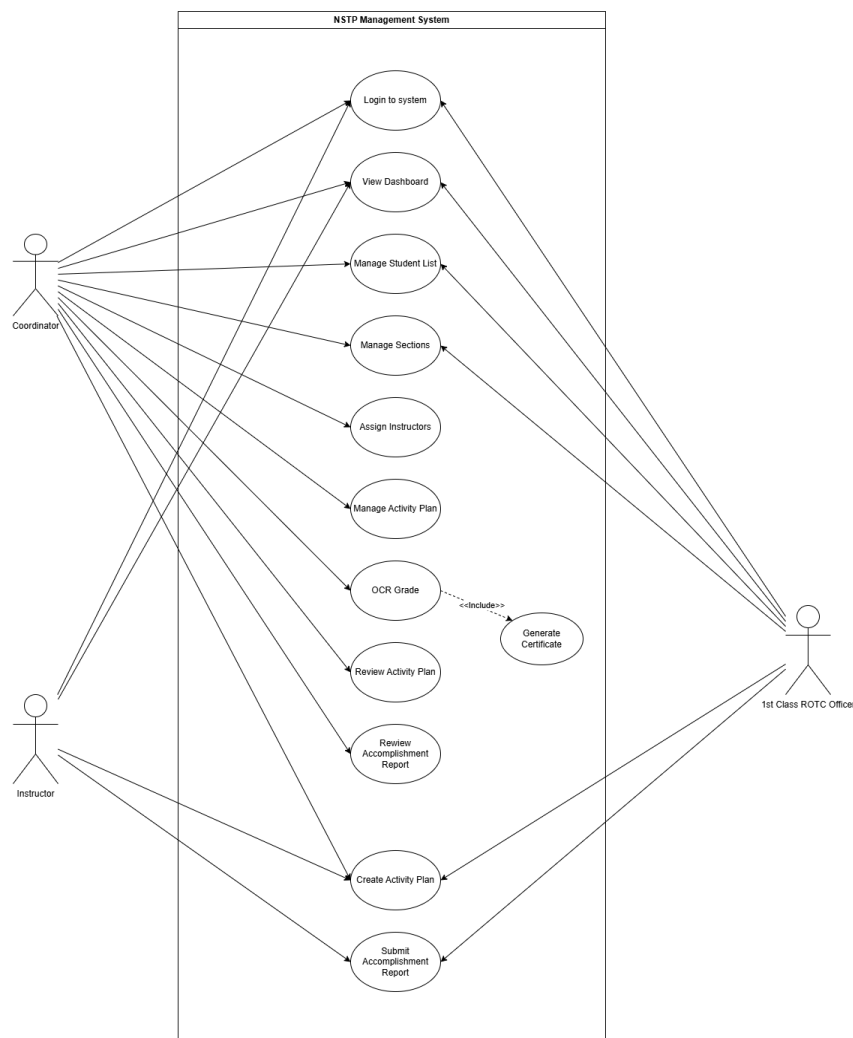
- **User Authentication:** Coordinator, Instructors and 1st Class ROTC Officers must have different access points to the system.
- **Student & Section Management:** Students should be able to be organized into sections of NSTP (CWTS, LTS, ROTC), assigned platoons, and to manage section lists.
- **Automated Grade Processing (OCR):** This system should be able to scan uploaded grade sheets, validate student grades (marking), and automatically classify students as "Passed" or "Failed".
- **Report & Document Workflow:** Instructors must be able to report on plans of activities and accomplishment reports, and the Coordinator must be able to review, approve, and/or reject.
- **Certificate Generation:** The system shall automatically generate downloadable and printable NSTP Completion Certificates based on the grades processed and upload of CHED serial numbers using OCR.
- **Audit Logging:** It should be able to track every user interaction, including data changes and report submissions, for transparency and security.

Non-Functional Requirements:

- **Security:** Sensitive student data (e.g., serial numbers) should be protected from unauthorized access, and user passwords should be protected.
- **Reliability:** Ensure that the system can process the OCR data reliably with minimal errors, to ensure the integrity of student records.
- **Usability:** The interface should be simple and easy to use to enable DNSC employees and instructors to navigate through the modules with minimal training.
- **Integrity:** The system should not allow further manipulation of the data after a report has been accepted or endorsed by the Coordinator.
- **Availability:** The web hosting platform must be available at all times, so that teachers can post reports when it is convenient.

Process Model: Use Case Diagram and Description

The Use Case Diagram of the NSTP Management System gives a good overview of how various users of the system will interact with the system and what specific features each of them will have. It graphically illustrates the major actors in the system, the Coordinator, Instructor and the 1st Class ROTC Officer and the tasks and responsibility assigned to each of the users. This diagram helps to define the basic processes in the system and they are, managing student lists, processing section assignments, making and revising activity plans, encoding grades and generating certificates. It also shows how each user gets into the system through a process known as a login process before performing the job he/she is assigned.



The NSTP Management System Use Case Description Table provides a listing of the various functions and interactions of the users in the system. It outlines the duties of the NSTP Coordinator, Instructors, and 1st Class ROTC Officers in NSTP records, NSTP sections, activity plans, accomplishment reports, grade processing, and generation of certificates. How the system aids in the secure management of data, grade processing by OCR, tracking of NSTP activities and the production of reports and completion certificates are also explained in the table.

Table X. Use Case Description for Registration

Use Case Name	Login to System
Participating Actors	• Coordinator • Instructor • 1st Class ROTC Officer
Entry Conditions	The user already has a registered account.
Flow of Events	1. The user opens the login page. 2. The user enters their email and passwords. 3. The user clicks the login button. 4. The system validates the credentials. 5. If valid, the system grants access and redirects to the dashboard.
Exit Conditions	The user successfully login to the system.
Quality Requirements	The system must ensure secure authentication and protect user credentials.

Use Case Name	View Dashboard
Participating Actors	• Coordinator • Instructor • 1st Class ROTC Officer
Entry Conditions	The user has already logged into the sytem
Flow of Events	1. The user logs into the system. 2. The system displays the dashboard. 3. The user views summarized data and system information.

	4. The system displays analytics such as overall students, performance and comparing the previous student population. 5. The system displays notification 6. The system shows the announcement board, calendar of activities for office-wide updates,
Exit Conditions	The dashboard is successfully displayed.
Quality Requirements	The system must display accurate and real time information.

Use Case Name	Manage Student List
Participating Actors	• ROTC Coordinator • 1st Class Officer
Entry Conditions	The coordinator and 1st class officer is already logged into the system
Flow of Events	1.The coordinator/ 1st class officer selects the sections module. 2.The system displays all the information of students such as course, year, components of NSTP. 3.The coordinator/1st class officer assigns, creates, edits, updates and deletes sections and platoon names. 4.For ROTC components, the 1st class officer organizes students into platoon lists. 5.The system validates the inputs and ensures no duplication or conflict. 6.The system saves the changes.
Exit Conditions	Sections are successfully created or updated.
Quality Requirements	The system must ensure proper organization and prevent duplicate sections.

Use Case Name	Manage Sections
Participating Actors	• Coordinator • 1st Class ROTC Officer
Entry Conditions	The coordinator and 1st Class ROTC Officer is already logged in to the system.

Flow of Events	1. The coordinator/1st Class ROTC Officer selects the sections module. 2. The system displays existing sections along with assigned student lists and ROTC platoons.. 3. The coordinator/1st Class ROTC Officer creates, edits, or deletes sections. 4. The coordinator/1st Class ROTC Officer assigns or updates section details, student information, and platoon names. 5. The coordinator/1st Class ROTC Officer assigns or updates the student list for each section. 6. For ROTC components, the 1st Class ROTC Officer organizes students into platoon lists. 7. The system validates the inputs and ensures no duplication or conflict. 8. The system saves the changes.
Exit Conditions	Sections are successfully created or updated.
Quality Requirements	The system must ensure proper organization and prevent duplicate sections.

Use Case Name	Assign instructors
Participating Actors	• Coordinator
Entry Conditions	The coordinator is logged into the system and sections already exist.
Flow of Events	1. The coordinator selects the instructor assignment module. 2. The system displays available instructors and NSTP sections, schedules and rooms. 3. The coordinator assigns instructors or ROTC officers to sections, schedules and rooms. 4. The system validates assignments. 5. The system saves the instructor assignments.
Exit Conditions	Instructors are successfully assigned to NSTP sections.
Quality Requirements	The system must prevent duplicate instructor assignments and ensure proper section allocation.

Use Case Name	Manage Activity Plan
Participating Actors	Coordinator
Entry Conditions	The coordinator is logged into the system.
Flow of Events	1: Coordinator chooses the activity plan management module. 2. The system shows submitted activity plans. 3. Coordinator reviews, edits, approves or rejects activity plans. 4. The status of the activity plan changes.
Exit Conditions	Activity plans are successfully managed and updated.
Quality Requirements	The system must securely store activity plans and maintain accurate approval records. .

Use Case Name	OCR Grade
Participating Actors	Coordinator
Entry Conditions	Grade sheets are uploaded into the system.
Flow of Events	1.The NSTP Coordinator uploads the NSTP grade sheets to the system. 2.The system uses OCR (Optical Character Recognition) technology to scan and read the uploaded files. 3. The system collects and verifies student grades. 4. The system will put students in Passed or Failed according to the grades. 5. A list of students that have successfully completed the NSTP program is produced. 6. The NSTP Coordinator e-mail the list of students who passed to CHED. 7. CHED issued official serial numbers for qualified students. 8. The system saves the grades and student status in the database.
Exit Conditions	Student grades are successfully processed, qualified students are identified, and the passed student list is prepared for CHED serial number processing.
Quality Requirements	The system must ensure accurate OCR processing, proper student classification, and reliable generation of passed student records.

Use Case Name	Generate Certificate
Participating Actors	Coordinator
Entry Conditions	Student grades are already processed, students are classified as passed, and CHED serial numbers are available in the system.
Flow of Events	1. The NSTP Coordinator selects the certificate generation module. 2. The coordinator uploads the pdf of CHED that contains Serial Number of students. 3. In the student list, a certificate icon appears next to a student's name only after their grade is processed and the CHED serial number is matched. 4. The system should show a status Certified label on the student's profile, where the Coordinator can click a print button to pop up the PDF. 5. The system stores the generated certificates in the database. 6. The NSTP Coordinator views, downloads, or prints the certificates.
Exit Conditions	NSTP completion certificates are successfully generated and stored.
Quality Requirements	The system must generate accurate certificates with complete student information and official CHED serial numbers.

Use Case Name	Review Activity Plan
Participating Actors	Coordinator
Entry Conditions	Submitted activity plans are available in the system.
Flow of Events	1. The coordinator selects the review activity plan module. 2. The system displays submitted activity plans. 3. The coordinator reviews the activity details.

	4. The coordinator approves or rejects the activity plan. 5. The system updates the activity plan status.
Exit Conditions	The activity plan review is completed.
Quality Requirements	The system must maintain accurate approval and review records.

Use Case Name	Review Accomplishment Report
Participating Actors	• Instructor • 1st Class ROTC Officer
Entry Conditions	The user already has a registered account.
Flow of Events	1.The instructor/1st Class ROTC Officer selects the accomplishment report module. 2. The system displays the accomplishment report form. 3. The instructor/1st Class ROTC Officer enters report details and uploads supporting documents. 4. The instructor/1st Class ROTC Officer submits the report. 5. The system records the submitted report.
Exit Conditions	The accomplishment report is successfully submitted.
Quality Requirements	The system must securely store submitted reports and attached files.

Use Case Name	Create Activity Plan
Participating Actors	• Instructor • 1st Class ROTC Officer
Entry Conditions	The instructor or 1st Class ROTC Officer is logged into the system.
Flow of Events	

	1. The instructor/1st Class ROTC Officer selects the create activity plan module. 2. The system displays the activity plan form. 3. The instructor/1st Class ROTC Officer enters activity details. 4. The instructor/1st Class ROTC Officer submits the activity plan. 5. The system records and forwards the activity plan for review.
Exit Conditions	The activity plan is successfully submitted.
Quality Requirements	The system must ensure complete and accurate activity plan submission.

Use Case Name	Submit Accomplishment Report
Participating Actors	• Instructor • 1st Class ROTC Officer
Entry Conditions	The activity is already complete and to be sent to the coordinator.
Flow of Events	1. The instructor/1st Class ROTC Officer selects the accomplishment report module. 2. The system displays the accomplishment report form. 3. The instructor/1st Class ROTC Officer enters report details and uploads supporting documents. 4. The instructor/1st Class ROTC Officer submits the report. 5. The system records the submitted report.
Exit Conditions	The accomplishment report is successfully submitted.
Quality Requirements	The system must securely store submitted reports and attached files.

Use Case Name	Archiving , calendar asa na under? , ang report and analytics asa ma under?, the audit logs?
---------------	--

Participating Actors	• Coordinator • Instructor • 1st Class ROTC Officer
Entry Conditions	The user already has a registered account.
Flow of Events	6. The user opens the login page. 7. The user enters their email and passwords. 8. The user clicks the login button. 9. The system validates the credentials. 10. If valid, the system grants access and redirects to the dashboard.
Exit Conditions	The user successfully login to the system.
Quality Requirements	The system must ensure secure authentication and protect user credentials.

(Add more tables based on your use case name)

Process Model: Activity Diagram

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(Insert Name of the Process here)

Figure X Lorem ipsum dolor sit amet consectetur adipiscing elit. Quisque faucibus ex sapien vitae pellentesque sem placerat. In id cursus mi pretium tellus duis convallis. Tempus leo eu aenean sed diam urna tempor. Pulvinar vivamus fringilla lacus nec metus bibendum egestas. laculis massa nisl malesuada lacinia integer nunc

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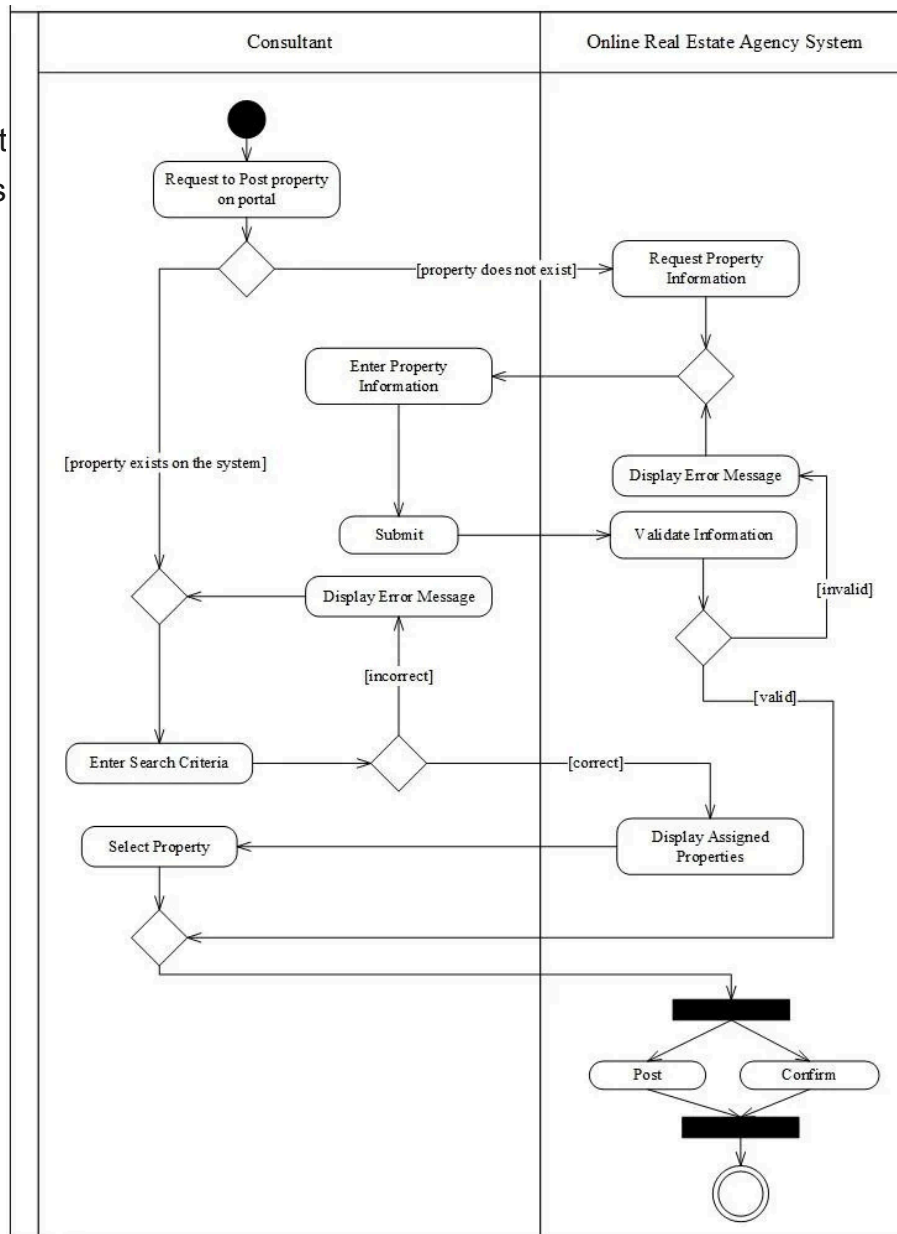


Figure X. Online Process Activity Diagram

(Disclaimer: This diagram is for sample purposes only and should not be the sole basis for your analysis. Please review previous courses and visit the library for additional references and comprehensive materials.)

(Add more diagrams based on the process)

Data Model: Context Data Flow Diagram

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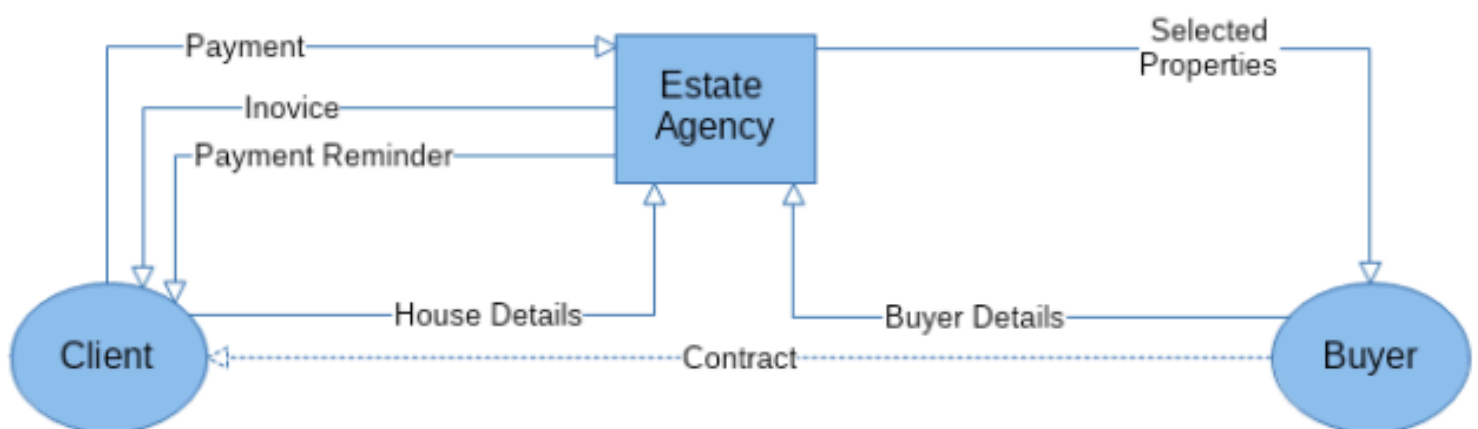


Figure X. Context Data Flow Diagram

(Disclaimer: This diagram is for sample purposes only and should not be the sole basis for your analysis. Please review previous courses and visit the library for additional references and comprehensive materials.)

Data Model: Level 1 Data Flow Diagram

The Level 1 data flow diagram for the system is Lorem ipsum dolor sit amet consectetur adipiscing elit. Quisque faucibus ex sapien vitae pellentesque sem placerat. In id cursus mi pretium tellus dui convallis. Tempus leo eu aenean sed diam urna tempor. Pulvinar vivamus fringilla lacus nec metus bibendum egestas. laculis massa nisl malesuada lacinia integer nunc posuere. Ut hendrerit semper vel class aptent taciti sociosqu. Ad litora torquent per conubia nostra inceptos himenaeos.



C. System Design

a. Database Design: Entity-Relationship Diagram

The Entity-Relationship Diagram (ERD) Lorem ipsum dolor sit amet consectetur adipiscing elit. Quisque faucibus ex sapien vitae pellentesque sem placerat. In id cursus mi pretium tellus dui convallis. Tempus leo eu aenean sed diam urna tempor. Pulvinar vivamus fringilla lacus nec metus bibendum egestas. laculis massa nisl malesuada lacinia integer nunc posuere. Ut hendrerit semper vel class aptent taciti sociosqu. Ad litora torquent per conubia nostra inceptos himenaeos.

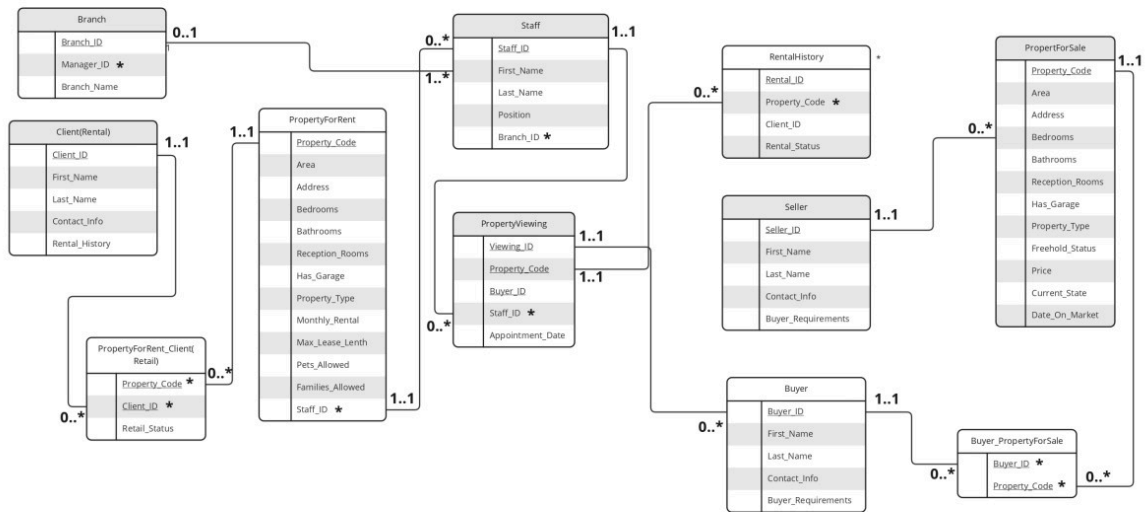


Figure X.Entity-Relationship Diagram

(Disclaimer: This diagram is for sample purposes only and should not be the sole basis for your analysis. Please review previous courses and visit the library for additional references and comprehensive materials.)

b. Data Dictionary

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Table X. Data Dictionary

Field Name	Data Type	Field Length	Constraint	Description
booking_id	int	11	PRIMARY KEY	Unique ID for the booking
start_date	date		NOT NULL	Booking's check-in date
end_date	date		NOT NULL	Booking's check-out date
start_time	time		NOT NULL	Booking's check-in time
end_time	time		NOT NULL	Booking's check-out time
school_year	varchar	20	NULL	Academic year
semester	varchar	10	NULL	Academic semester
organization	varchar	255	NULL	Organization making the booking

contact_info	varchar	255	NULL	Contact information
purpose	varchar	255	NULL	Purpose of the booking
num_users	int	11	NULL	Number of users
equipment_needed	text		NOT NULL	Equipment needed for the booking
activity_design	Varchar	255	NOT NULL	Activity plan requirement
request_letter	varchar	255	NOT NULL	Request Letter contents
status	varchar	20	NULL	Status of the booking
created_at	timestamp		NOT NULL	Record was created

(Add more tables based on your entity mentioned in the ERD)

c. System Architecture

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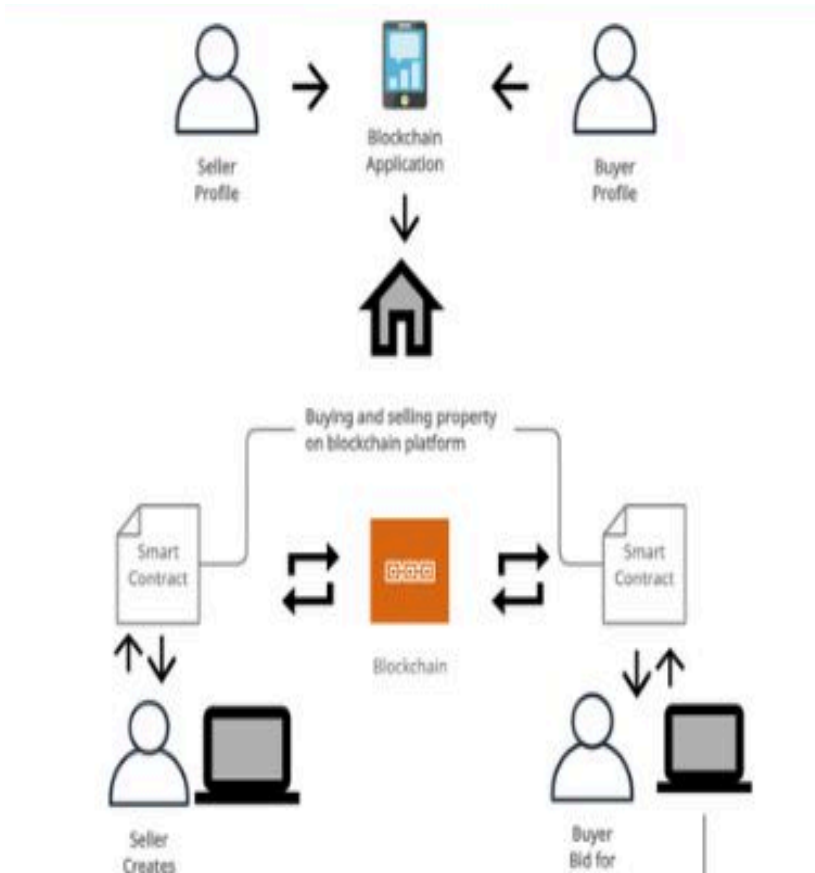


Figure X. System Architecture

(Disclaimer: This diagram is for sample purposes only and should not be the sole basis for your analysis. Please review previous courses and visit the library for additional references and comprehensive materials.)

D. System development and deployment

a. Test Cases, Alpha Testing, and UAT

The system will go through strict testing to ensure it has no errors. First, the Programmer will perform Alpha Testing to check for broken code and database issues. Then, the NSTP Coordinator and Instructors will perform User Acceptance Testing (UAT). They will try to upload grades and create certificates to prove the system handles real-world school tasks perfectly.

b. Deployment Plan with Sampling Technique

The system will be deployed in phases. We will use a "Pilot Testing" sampling technique. This means the system will first be used by only one ROTC platoon and one CWTS section. If successful, it will be rolled out to all NSTP students across the whole college.

c. Questionnaire

After deployment, users will fill out a survey based on the ISO 25010 software quality standards. The questionnaire will ask the NSTP staff and students to rate the system's speed, ease of use, visual design, and reliability on a scale of 1 to 5.

E. System Maintenance

a. Risk Management Plan

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Table X. Risk Management Plan for the (Name of your system)

Risk ID	Risk	Description	Likelihood	Severity	Risk Factor	Actions to Minimize Risk
1	Internal Risks					
1.1						

1.2						
1.3						
1.4						
1.5						
2	External Risks					
2.1						
2.2						
2.3						
2.4						
2.5						

b. Project Turnover Plan

Once the system is fully working, the project will be officially handed over to the DNSC NSTP Office. The team will provide a User Manual and conduct a half-day training seminar for the coordinators and instructors. Passwords and admin access will be transferred to the official school IT personnel.

Table X. Project Closure Report

PROJECT CLOSURE REPORT				
Project Name	NATIONAL SERVICE TRAINING PROGRAM MANAGEMENT SYSTEM FOR DAVAO DEL NORTE STATE COLLEGE			
Brief Description	A web-based platform to automate NSTP student records, OCR grade uploads, and certificate generation.			
Project Sponsor	DNSC NSTP Coordinator			
Prepared by	Project Team			
Objectives Achievement				
Objectives	Description	Performance	Achieved	Not Achieved
CENTRALIZED RECORDS	Store all CWTS, LTS, and ROTC			

	student data in a database			
OCR GRADE UPLOADS	Allow coordinator to scan grade sheets instead of manually checking and typing them			
AUTOMATED CERTIFICATES	Generate completion certificates automatically using official CHED serial numbers			
<i>Approvals</i> This document requires the approval:				
Position	Signature		Date	
Project Manager				
Project Sponsor				